R10.R

r3914162

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```
library(readxl)
carga = read_excel("carga_no_ponto_de_falha.xlsx")
carga = as.numeric(carga$Carga)
# 1)
t.test(carga,
       mu = 10,
       alternative = "greater",
       conf.level = 0.95)
```

```
##
##  One Sample t-test
##
## data:  carga
## t = 4.9017, df = 21, p-value = 3.781e-05
## alternative hypothesis: true mean is greater than 10
## 95 percent confidence interval:
##  12.40996      Inf
## sample estimates:
## mean of x
##  13.71364
```

```
# 2)
catalisadores = read_excel("catalisadores.xlsx")
cata1 = as.numeric(catalisadores$catalisador_1)
cata2 = as.numeric(catalisadores$catalisador_2)

t.test(cata1, cata2,
          var.equal = TRUE,
          alternative = "two.sided",
          conf.level = 0.95)
```

```
##
##  Two Sample t-test
##
## data:  cata1 and cata2
## t = -0.35359, df = 14, p-value = 0.7289
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -3.373886  2.418886
## sample estimates:
## mean of x mean of y
##   92.2550   92.7325
```

```r
# 4)
shapiro.test(carga)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  carga
## W = 0.96981, p-value = 0.7067
```

```r
# 5)
shapiro.test(cata1)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  cata1
## W = 0.92171, p-value = 0.4439
```

```r
shapiro.test(cata2)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  cata2
## W = 0.88182, p-value = 0.196
```